

AI in sociological research

Some ways I think we can use these tools
in our own research

Last year's discussion focused on student responses and essays.
Today I want to look at what happens inside a research project.

[Open the demonstration files](#)

Cognitive offloading and learning

External resources change what we need to remember and do ourselves.

Trends in Cognitive Sciences

CellPress

Review

Cognitive Offloading

Evan F. Risko^{1,*} and Sam J. Gilbert^{2,*}

2016 · Review

Risko & Gilbert

Offloading reduces cognitive demands. Consequences depend on the task and use of the aid.

The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects From a Survey of Knowledge Workers

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2025 · Workplace survey

Lee et al.

319 workers: confidence in AI was associated with less reported critical thinking.

PNAS

RESEARCH ARTICLE

ECONOMIC SCIENCES

Generative AI without guardrails can harm learning: Evidence from high school mathematics

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Affiliations are included on p. 7.

Edited by Emma Brunskill, Stanford University, Stanford, CA; received November 3, 2024; accepted May 5, 2025 by Editorial Board Member Mark Granovetter

2025 · Randomized school trial

Bastani et al.

Unaided exam grades fell after unrestricted AI practice help. A guided tutor largely removed the penalty.

Reduced effort, reduced learning and lasting cognitive loss are different claims.

[Actual paper excerpts · click a screenshot to open its source](#)

What research agents can accomplish

Article

Accelerating scientific discovery with Co-Scientist

<https://doi.org/10.1038/s41586-026-10644-y>

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Check for updates

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Google · Nature, 2026

Co-Scientist

Agents proposed hypotheses; scientists selected and tested them in the lab.

A system can extend what a researcher can execute while still needing their judgment.

[METR: what a time horizon measures](#)

Actual paper excerpts · selected studies, not a head-to-head comparison.

Can AI agents conduct open-ended AI research?

Early evidence from two case studies

Peter Kirgis^{1,*}, Sayash Kapoor^{1,*}, Andrew Schwartz^{1,2}, Stephan Rabanser¹, David Africa³, Konstantinos Voudouris³, Viet Nguyen⁴, Toby Pilditch³, Magda Dubois³, Harry Coppock³, Cozmin Ududec³, Nitya Nadgir⁵, Matilda Orona⁶, Tilman Bayer⁵, Derrick Chan-Sew⁵, Yue Ling⁵, Abhishek Shetty⁵, Helen Toner⁷, Gillian Hadfield⁸, Seth Lazar⁸, Steve Newman⁹, Shoshannah Tekofsky¹⁰, Rishi Bommasani¹¹, Arvind Narayanan¹

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Kirgis et al., including Narayanan · 2026 preprint

Two open-ended research tasks

Six days of research: engineering completed, but both papers judged below the publication bar.

2025: the chat window

Illustrated exchange

Summarize research on chatbots and support-seeking from friends and family.

A prose response appears here, with suggested themes and references to inspect.

Ask a follow-up...

The researcher carries the workflow

Read the answer.

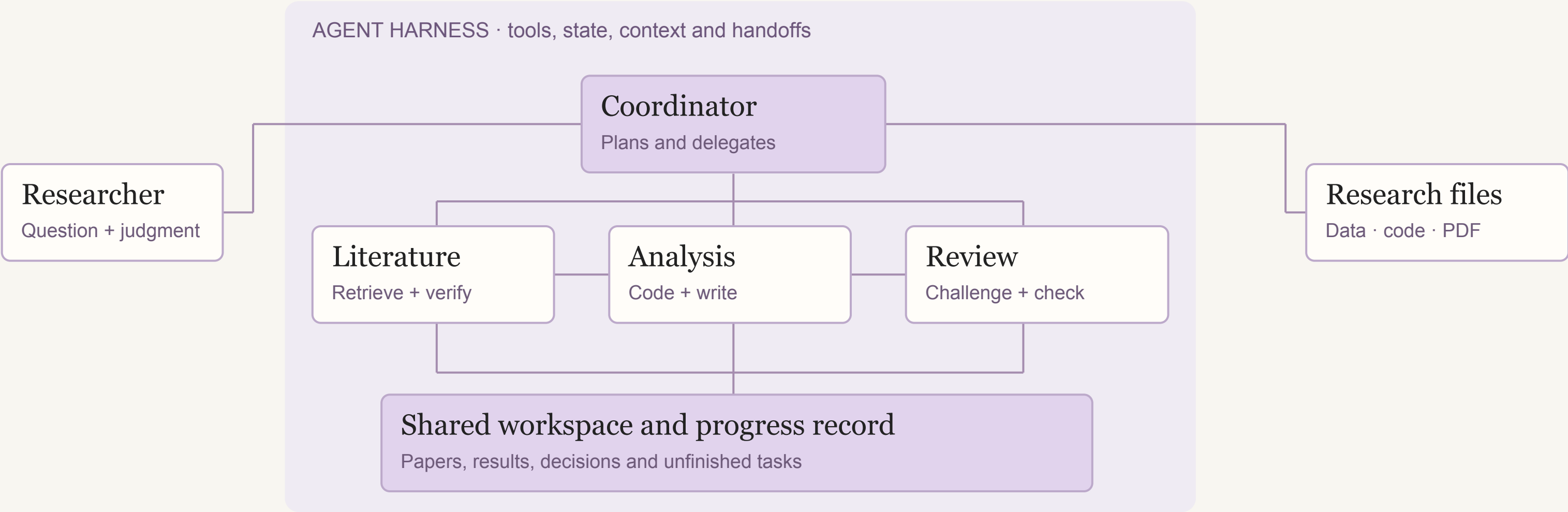
Find the papers.

Move material into files.

Run analysis elsewhere.

The use case discussed last year. Tool-using and multi-agent systems already existed in 2025.

2026: working with a team of agents



The harness manages execution around the model. The researcher remains part of the revision loop.

[Open the four-stage interactive diagram](#)

Teaching schematic; no agents are running on the slide.

1. Complete cognitive offloading

- Give it a bounded task and inspect what comes back.
- “Reproduce this figure from the supplied data and document any discrepancies.”
- Other uses: bibliographic records, file conversion, replication checks.

The failure is accepting an output because it looks finished.

2. Guided collaboration

- Let it do much of the work; intervene where a decision matters.
- “Show me the proposed measures before running the analysis.”
- Other uses: developing an analysis, revising a coding scheme, comparing operationalizations.

The failure is approving each step without really examining it.

3. The pain-in-the-ass coauthor

You bring an argument. It questions it. You respond, and work through the implications together.

“I think these findings support displacement. Here is why.
What am I missing?”

Useful for theory, measurement, identification and interpretation.
Its objections need scrutiny too; it can misunderstand the evidence.

Do chatbots change how people seek support?

Heavy ChatGPT users tend to be more lonely, suggests research

Rachel Hall, The Guardian, 25 March 2025

Our question: do AI chatbots reduce requests for support from friends and family?

[Open the article](#)

What would you actually type?

Open a project folder, add the skill, then give it the question.

Codex CLI / IDE: `$literature-audit`

Claude Code: `/literature-audit`

“Find primary studies. Check the outcome instruments and source versions. Write an evidence table and bibliography. Ask me about consequential ambiguities.”

[Open the skill and setup instructions](#)

Workshop-authored skill; search and file tools depend on the session.

Inspect the evidence table and retrieval record

The agent assembled nine records. The useful part is seeing what each study actually measured.

Study design	What was assigned, and what was self-selected?
Outcome	Requests, socialization, loneliness, or response ratings?
Source check	Which version? Full text or abstract only?

[Open evidence table](#)

[Open retrieval record](#)

[Open Fang et al., v2](#)

Working through a measurement decision

RESEARCHER

“Could we compare the share of support needs that lead to a friend?”

AI COAUTHOR

“What if chatbot use changes what people report as a need?
Then the denominator changes too. Could total requests over
a fixed follow-up be the primary outcome?”

[Open the full exchange](#)

Scripted teaching example; the actual reviewer reports are linked separately.

Two agents checked the proposal

- The source reviewer checked the claim against the paper.
- The methods reviewer examined the outcome, comparison and analysis.
- They then read each other's objections. One unresolved issue: a contact-prompt effect does not establish a bridging mechanism.

[Open the actual reports and cross-checks](#)

The resulting research proposal

A short protocol with an evidence assessment, defined outcome, assignment scheme, analysis and draft diary questions.

- Compare a standard chatbot offer with usual access.
- Separately compare an added human-contact feature with the standard offer.
- Retain non-users in assigned groups; missing diaries are not zero requests.

[Open the compiled PDF](#)

[Open LaTeX](#)

[Open bibliography](#)